C Programming – Assignment 01

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Program Solutions

Membership input for Question 1 accepts true/false or 1/0. Assumptions are noted where the assignment statement is ambiguous.

Question 1: Cab Fare Calculation

```
/* Cab Fare Calculation - C Programming Assignment 01 */
#include <stdio.h>
#include <string.h>

int main(void) {
    double distance, fare = 100.0;
    char member[10];

    scanf("%lf", &distance);
    scanf("%9s", member); /* Accept true/false or 1/0 */

    if (distance > 10.0) {
        fare += (distance - 10.0) * 20.0;
    }

    if (strcmp(member, "true") == 0 || strcmp(member, "TRUE") == 0 ||
        strcmp(member, "1") == 0) {
        fare *= 0.90;
    }

    printf("Final Fare: ₹%.0f\n", fare);
    return 0;
}
```

Note: Input distance, then membership as true/false or 1/0.

Question 2: Online Examination System

```c
/* Online Examination System - C Programming Assignment 01 */
#include <stdio.h>

int main(void) {
    int questionsAnswered, remainingTime, alreadySubmitted;

    scanf("%d", &questionsAnswered);
    scanf("%d", &remainingTime);
    scanf("%d", &alreadySubmitted);

    if (questionsAnswered >= 1 && remainingTime > 0 && alreadySubmitted == 0) {
        printf("Submission Allowed\n");
    } else {
        printf("Submission Not Allowed\n");
    }

    return 0;
}
```

Note: Input questions answered, remaining time, and submitted status (1=yes, 0=no).

Question 3: Treasure Hunt

```c
/* Treasure Hunt - C Programming Assignment 01 */
#include <stdio.h>

int main(void) {
    int treasure, x, y;
    int longBen, blackbeard, remaining, eachPirate;

    scanf("%d", &treasure);
    scanf("%d", &x);
    scanf("%d", &y);

    longBen = treasure * x / 100;
    remaining = treasure - longBen;
    blackbeard = remaining * y / 100;
    remaining -= blackbeard;
    eachPirate = remaining / 3;

    printf("%d\n%d\n%d\n", longBen, blackbeard, eachPirate);
    return 0;
}
```

Note: Integer division discards fractional coins; the remaining coins are split among three pirates.

Question 4: Dollars and Cents

```c
/* Dollars and Cents - C Programming Assignment 01 */
#include <stdio.h>

int main(void) {
    int dollars1, cents1, dollars2, cents2;
    int totalCents, dollars, cents;

    scanf("%d%d%d%d", &dollars1, &cents1, &dollars2, &cents2);

    totalCents = dollars1 * 100 + cents1 + dollars2 * 100 + cents2;
    dollars = totalCents / 100;
    cents = totalCents % 100;

    printf("%d\n%d\n", dollars, cents);
    return 0;
}
```

Note: Input dollar and cent values for both amounts.

Question 5: Car Mileage

```
/* Car Mileage - C Programming Assignment 01 */
#include <stdio.h>

int main(void) {
    float mileage;
    int petrolLitres, distance;

    scanf("%f%d%d", &mileage, &petrolLitres, &distance);

    if (mileage * petrolLitres >= distance) {
        printf("Can reach\n");
    } else {
        printf("Cannot reach\n");
    }

    return 0;
}
```

Note: Input mileage (km/litre), petrol litres, and remaining distance (km).

Question 6: Car Race

```c
/* Car Race - C Programming Assignment 01 */
#include <stdio.h>

int main(void) {
    int a, b, c, carNumber;

    scanf("%d%d%d%d", &a, &b, &c, &carNumber);

    if (a % carNumber == 0) {
        printf("Car %d goes into road A\n", carNumber);
    } else if (b % carNumber == 0) {
        printf("Car %d goes into road B\n", carNumber);
    } else if (c % carNumber == 0) {
        printf("Car %d goes into road C\n", carNumber);
    } else {
        printf("No path exists\n");
    }

    return 0;
}
```

Note: Assumption from sample: car number must divide the road number; first matching road is selected.

Question 7: Debt Repay

```c
/* Debt Repay - C Programming Assignment 01 */
#include <stdio.h>

int main(void) {
    double principal, rate, years;
    double interest, amount, discount, settlement;

    scanf("%lf%lf%lf", &principal, &rate, &years);

    interest = principal * rate * years / 100.0;
    amount = principal + interest;
    discount = interest * 0.02;
    settlement = amount - discount;

    printf("%.2f\n%.2f\n%.2f\n%.2f\n",
           interest, amount, discount, settlement);

    return 0;
}
```

Note: Uses simple interest; 2% discount is applied to the interest.

Question 8: Employee Job Level

```c
/* Employee Job Level - C Programming Assignment 01 */
#include <stdio.h>

int main(void) {
    int experience, performance;

    scanf("%d%d", &experience, &performance);

    if (experience < 3) {
        printf("Level: Junior\n");
    } else if (experience <= 7) {
        if (performance >= 80) {
            printf("Level: Mid-Level\n");
        } else {
            printf("Level: Junior\n");
        }
    } else {
        if (performance >= 85) {
            printf("Level: Senior\n");
        } else {
            printf("Level: Mid-Level\n");
        }
    }

    return 0;
}
```

Note: For unspecified below-threshold performance cases, the next lower level is used.

Question 9: Number of Days

```
/* Number of Days - C Programming Assignment 01 */
#include <stdio.h>

int main(void) {
    int year, month, days;

    scanf("%d%d", &year, &month);

    if (year < 1900 || year > 9999 || month < 1 || month > 12) {
        printf("0\n");
        return 0;
    }

    switch (month) {
        case 4: case 6: case 9: case 11:
            days = 30;
            break;
        case 2:
            if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0)) {
                days = 29;
            } else {
                days = 28;
            }
            break;
        default:
            days = 31;
    }

    printf("%d Days\n", days);
    return 0;
}
```

Note: Invalid year or month prints 0; leap years are handled.

Question 10: Electricity Bill

```c
/* Electricity Bill - C Programming Assignment 01 */
#include <stdio.h>

int main(void) {
    int units;
    double bill;

    scanf("%d", &units);

    if (units < 0) {
        printf("Invalid units\n");
    } else if (units <= 200) {
        bill = units * 0.50;
        printf("Rs.%.0f\n", bill);
    } else if (units <= 400) {
        bill = units * 0.65 + 100;
        printf("Rs.%.2f\n", bill);
    } else if (units <= 600) {
        bill = units * 0.80 + 200;
        printf("Rs.%.2f\n", bill);
    } else {
        bill = units * 1.25 + 425;
        printf("Rs.%.2f\n", bill);
    }

    return 0;
}
```

Note: Uses the rate-times-total-units interpretation implied by the sample.